

Sim2 45° CLIP MCDO Noise Study

1. Introduction

We study how additive noise and progressive downsampling alter the geometry of CLIP image embeddings when Monte Carlo Dropout (MCDO) is applied to the vision tower. The goal is to quantify how perturbations change the stochastic embedding cloud (variance, orientation, and mean drift) so we can anticipate failure modes in downstream retrieval or classification settings. All analyses focus on embedding-space diagnostics rather than classifier accuracy, mirroring a scientific report structure: we document the experimental configuration, define uncertainty metrics, and then interpret how those metrics respond to perturbations of increasing severity.

2. Methodology

- **Dataset:** data/car_sim/sim2_cropped_45deg
- **Model:** openai/clip-vit-base-patch32
- **Dropout p:** 0.01
- **Passes:** 64
- **Images:** 46
- **Dropout instrumentation:** DropoutAdapter wraps all twelve ViT encoder blocks plus the visual projection head ($p = 0.01$).
- **Sampling:** 64 stochastic passes per image (microbatch 4, deterministic seed). Deterministic baselines use a single pass with dropout disabled.
- **Perturbation sweep:** Gaussian noise $\sigma \in \{0.01, \dots, 0.5\}$ and downsampling up to 93% ($224 \rightarrow 16$ px) followed by bicubic upsampling.
- **Pass-stability sweep:** Additional runs at $T \in \{2, 4, 8, 16, 32, 64, 128\}$ for original, strongest-noise, and strongest-downsampling cases.

3. Metric Primer

- **Trace ($\text{Tr } \Sigma$):** Sum of covariance eigenvalues; measures total dispersion of the embedding cloud. Increases imply broader uncertainty.
- **Log-determinant ($\log \det \Sigma$):** Log-volume of the covariance ellipsoid. It captures how uncertainty spreads across dimensions; drops signal collapse into a lower-dimensional subspace.
- **Off-diagonal mass:** L_1 magnitude of non-diagonal covariance entries; indicates cross-dimensional coupling and anisotropy.
- **Mean shift / Mahalanobis shift:** L_2 distance and covariance-normalised distance between the stochastic mean and the deterministic embedding, quantifying drift induced by perturbations.
- **Tangent trace / $\lambda_{\max}$:** Variance within the hypersphere orthogonal to the mean direction; highlights changes in directional uncertainty.
- **Spectral entropy & top-10 share:** Describe how variance concentrates across eigenmodes. Lower entropy or higher top-10 share means uncertainty is dominated by a few directions.
- **Circular variance:** Dispersion of unit-normalised samples; complements tangent analysis by examining orientation consistency.
- **Pass-count stability:** How rapidly trace/logdet estimates converge as the number of stochastic passes T increases.

4. Aggregate MCDO Embedding Metrics

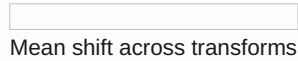
Baseline trace is 12.31, with logdet -6376.03. Noise and crop-based degradations broaden the covariance while logdet declines; we report anisotropy using correlation-based Frobenius norms for scale-free comparison.

Figure 1. Aggregate metrics overview. Aggressive downsampling emerges as the

dominant driver of higher trace and lower log-volume across transforms.



Figure 2. Mean shift across transforms. Mean embedding displacement mirrors the same ordering: raw (nearest) and smoothed (bicubic) crops introduce the largest drift from the deterministic baseline.



5. Deterministic Baseline (1 pass, dropout disabled)

With dropout disabled every run collapses to trace $\approx 5.12 \times 10^{-4}$ and a singular covariance; logdet and anisotropy are therefore undefined.

6. Sensitivity to Noise

Trace grows steadily with σ ; at $\sigma=0.50$ it reaches 12.80 ($\approx+4.0\%$ vs baseline) while logdet shifts to -6383.93 (-7.89 vs baseline), signalling that variance expands yet concentrates into fewer dominant axes.

Figure 3. Relative trace change under noise. Noise-only perturbations peak at +4.0% trace change when σ reaches its maximum setting.

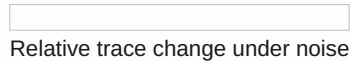


Figure 4. Logdet shift under noise. Log-volume steadily contracts with noise strength, finishing at -7.89 relative to baseline.

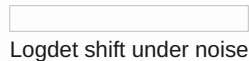


Figure 5. Trace vs noise severity. Trace increases monotonically with σ , indicating broader stochastic clouds as perturbations intensify.

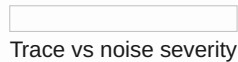


Figure 6. Logdet vs noise severity. Log-volume falls steadily with stronger noise, reflecting variance concentrating into fewer dominant modes.

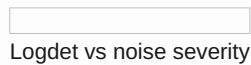


Figure 7. Correlation anisotropy vs noise severity. Cross-dimensional coupling grows gradually with σ , showing correlated drift rather than isotropic spread.

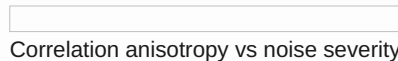
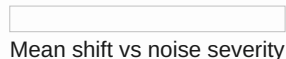


Figure 8. Mean shift vs noise severity. Monte Carlo means drift further from the deterministic embedding as σ increases, tracking the speckle strength.



7. Sensitivity to Downsampling

Smoothed crops (bicubic upsample) beyond 60% sharply increase trace (e.g., 93% reduction $\rightarrow$ trace 12.83) while logdet becomes non-monotone—after a mild

plateau it collapses to -6380.87 (volume ratio ≈ 0.008), consistent with aliasing and patch-token collapse once only a few pixels remain.

Figure 9. Relative trace change under smoothed crop (bicubic). Smoothed crops peak around +4.2% trace change once spatial resolution drops to 93%.



Relative trace change under smoothed crop (bicubic)

Figure 10. Logdet shift under smoothed crop (bicubic). Covariance volume contracts by -4.84 at the harshest smoothed crop, underscoring aliasing-driven collapse.



Logdet shift under smoothed crop (bicubic)

Figure 11. Trace distribution per transform. Trace variance fans out for the most aggressive downsampling settings, underscoring their broader uncertainty.



Trace distribution per transform

Figure 12. Trace vs downsampling severity. Trace rises almost linearly until about 60% reduction before the 93% case accelerates beyond +4.2%.



Trace vs downsampling severity

Figure 13. Logdet vs downsampling severity. Log-volume stays flat for mild reductions then plunges once images fall below 20% of their original side length.



Logdet vs downsampling severity

Figure 14. Correlation anisotropy vs downsampling severity. Severe downsampling amplifies cross-dimensional coupling, showing the embedding cloud stretching along fewer axes.



Correlation anisotropy vs downsampling severity

Figure 15. Mean shift vs smoothed crop severity. Mean displacement spikes once resolution falls below 60%, matching the trace surge caused by aggressive smoothing.



Mean shift vs smoothed crop severity

Figure 16. Logdet distribution per transform. The tail of extreme downsampling skews toward very low logdet values, reinforcing the volume collapse story.



Logdet distribution per transform

Figure 17. Off-diagonal mass distribution per transform. Off-diagonal covariance mass swells as resolution drops, highlighting stronger axis entanglement.



Off-diagonal mass distribution per transform

Raw crop (nearest)

Keeping the raw nearest-neighbour crop (no smoothing) produces larger block artefacts; the harshest setting (pixel_downsample_93pct) yields trace 12.37 while logdet drops to -6382.75, indicating strong concentration of variance into a handful of axes.

Figure 18. Relative trace change under raw crop (nearest). Trace rises steadily as the grid coarsens, confirming block artefacts inflate stochastic spread.

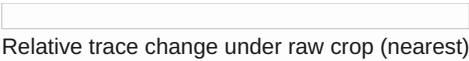


Figure 19. Relative logdet change under raw crop (nearest). Log-volume decays more sharply once smoothing is removed, highlighting how raw crops squeeze variance into fewer modes.

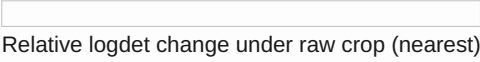


Figure 20. Trace vs raw crop severity. Nearest-neighbour crops reach +0.5% trace change once the short side is reduced by 93%.

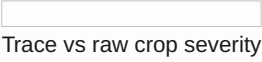


Figure 21. Logdet vs raw crop severity. Logdet collapses to -6382.75 (-6.71 vs baseline) for the 93% setting, showing volume loss once smoothing is removed.

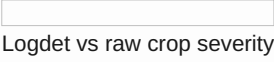


Figure 22. Correlation anisotropy vs raw crop severity. Raw crops amplify anisotropy faster than smoothed crops because hard edges align variance along token boundaries.

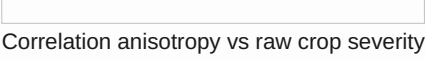


Figure 23. Mean shift vs raw crop severity. Nearest-neighbour reductions drive the largest centroid drift as aliasing artefacts dominate token activations.

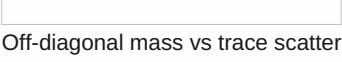


8. Cross-Metric Geometry

Figure 24. Logdet vs trace scatter. Smoothed crop points cluster in the high-trace, low-logdet corner, separating cleanly from the noise-induced shifts.



Figure 25. Off-diagonal mass vs trace scatter. Off-diagonal coupling grows hand-in-hand with trace for the strongest smoothed crops, reinforcing anisotropy concerns.



9. Mean Shift Diagnostics

transform	L2 shift	Mahalanobis shift
original	1.5934 ± 0.2906	806.9347 ± 97.6104
gaussian_noise_010	2.7813 ± 0.6508	1666.2401 ± 255.5203
gaussian_noise_020	3.6054 ± 0.6817	2227.3307 ± 358.3126
gaussian_noise_050	4.3841 ± 0.6772	2788.7280 ± 342.4279
gaussian_noise_100	5.1952 ± 0.9378	3367.2794 ± 419.5660
gaussian_noise_200	6.5398 ± 1.0501	4228.4310 ± 535.9685
gaussian_noise_500	8.7011 ± 0.8827	5537.6085 ± 582.0240
saltpepper_5pct	5.8839 ± 0.9645	3828.5899 ± 468.3757
downsample_1pct	2.9156 ± 0.7442	1763.3735 ± 465.4738
pixel_downsample_1pct	3.8731 ± 1.1712	2337.2949 ± 742.3540
downsample_2pct	2.9370 ± 0.7538	1767.1616 ± 482.2413
pixel_downsample_2pct	3.8726 ± 1.1516	2337.8885 ± 712.3581
downsample_5pct	3.0156 ± 0.7507	1832.2940 ± 465.8001
pixel_downsample_5pct	3.9265 ± 1.2046	2401.8979 ± 793.1497
downsample_10pct	3.1176 ± 0.7988	1908.4585 ± 494.0486
pixel_downsample_10pct	4.0919 ± 1.1790	2516.4781 ± 788.3843
downsample_20pct	3.3776 ± 0.8529	2091.0686 ± 520.2487
pixel_downsample_20pct	4.4315 ± 1.1498	2813.4398 ± 791.4507
downsample_40pct	4.0572 ± 0.8832	2561.1259 ± 518.2966
pixel_downsample_40pct	5.1926 ± 1.0419	3405.3319 ± 674.2687
downsample_60pct	5.1260 ± 0.8931	3215.4408 ± 481.4732
pixel_downsample_60pct	6.2757 ± 1.0407	4071.8815 ± 596.6683
downsample_80pct	6.9914 ± 0.7538	4419.8093 ± 534.7429
pixel_downsample_80pct	8.7442 ± 0.9495	5380.3463 ± 604.2322
downsample_85pct	7.7168 ± 0.6615	4915.8207 ± 549.4709
pixel_downsample_85pct	9.3134 ± 0.7255	5702.9129 ± 539.7879
downsample_90pct	8.8262 ± 0.6330	5610.9697 ± 584.7671
pixel_downsample_90pct	9.7077 ± 0.5933	6007.7963 ± 578.3062
downsample_93pct	9.3279 ± 0.6722	5948.2206 ± 621.6978
pixel_downsample_93pct	9.7685 ± 0.5089	6197.0443 ± 548.2255

Figure 26. Mean shift vs noise severity. Mean L2 displacement tracks noise strength, reinforcing that stochastic means drift progressively under σ increases.


Mean shift vs noise severity

Figure 27. Mean shift vs downsampling severity. Resolution loss beyond 60% triggers a rapid rise in mean shift, reflecting aliasing-induced drift.


Mean shift vs downsampling severity

Figure 28. Mean shift vs pixelated downsampling severity. Pixelation pushes mean drift even harder once large blocks emerge, underscoring the harsher aliasing penalty.


Mean shift vs pixelated downsampling severity

Figure 29. Mahalanobis shift vs noise severity. Covariance-normalised drift escalates with σ , showing that noise injects uncertainty aligned with high-variance directions.


Mahalanobis shift vs noise severity

Figure 30. Mahalanobis shift vs downsampling severity. Heavy downsampling rockets Mahalanobis distance, indicating the embedding cloud moves far relative to its contracted covariance.

Mahalanobis shift vs downsampling severity

Figure 31. Mahalanobis shift vs pixelated downsampling severity. Nearest-neighbour reductions yield the largest covariance-normalised drift, showing block artefacts distort embeddings most severely.

Mahalanobis shift vs pixelated downsampling severity

10. Spectral & Tangent Geometry

transform	tangent trace	tangent $\lambda_{\max}$	spectral entropy	top-10 share	circular variance
original	8.5239 $\pm$ 1.6899	2.2936 $\pm$ 0.9171	2.5811 $\pm$ 0.2383	0.7565 $\pm$ 0.0436	0.0633 $\pm$ 0.0157
gaussian_noise_010	8.7489 $\pm$ 1.7260	2.3569 $\pm$ 0.9274	2.5859 $\pm$ 0.2303	0.7557 $\pm$ 0.0424	0.0654 $\pm$ 0.0161
gaussian_noise_020	8.7992 $\pm$ 1.6985	2.3006 $\pm$ 0.9398	2.6201 $\pm$ 0.2280	0.7518 $\pm$ 0.0420	0.0631 $\pm$ 0.0155
gaussian_noise_050	8.5107 $\pm$ 1.6004	2.2645 $\pm$ 0.9120	2.5796 $\pm$ 0.2205	0.7574 $\pm$ 0.0400	0.0617 $\pm$ 0.0152
gaussian_noise_100	8.4447 $\pm$ 1.5670	2.2813 $\pm$ 0.9015	2.5460 $\pm$ 0.2187	0.7620 $\pm$ 0.0389	0.0625 $\pm$ 0.0152
gaussian_noise_200	8.2401 $\pm$ 1.3829	2.1899 $\pm$ 0.8128	2.5005 $\pm$ 0.2321	0.7685 $\pm$ 0.0389	0.0601 $\pm$ 0.0144
gaussian_noise_500	7.7364 $\pm$ 1.4081	2.0397 $\pm$ 0.7857	2.3501 $\pm$ 0.2266	0.7948 $\pm$ 0.0384	0.0552 $\pm$ 0.0146
saltpepper_5pct	8.6093 $\pm$ 1.3903	2.2036 $\pm$ 0.8332	2.5686 $\pm$ 0.2253	0.7608 $\pm$ 0.0370	0.0626 $\pm$ 0.0142
downsample_1pct	8.6512 $\pm$ 1.6931	2.3227 $\pm$ 0.9141	2.5873 $\pm$ 0.2266	0.7572 $\pm$ 0.0406	0.0619 $\pm$ 0.0148
pixel_downsample_1pct	8.4435 $\pm$ 1.5675	2.2622 $\pm$ 0.8377	2.5307 $\pm$ 0.2185	0.7668 $\pm$ 0.0402	0.0624 $\pm$ 0.0152
downsample_2pct	8.6612 $\pm$ 1.6923	2.3173 $\pm$ 0.9090	2.5889 $\pm$ 0.2245	0.7570 $\pm$ 0.0402	0.0620 $\pm$ 0.0149
pixel_downsample_2pct	8.4704 $\pm$ 1.5897	2.2443 $\pm$ 0.8321	2.5379 $\pm$ 0.2177	0.7661 $\pm$ 0.0397	0.0625 $\pm$ 0.0153
downsample_5pct	8.6939 $\pm$ 1.6944	2.3229 $\pm$ 0.9144	2.5880 $\pm$ 0.2230	0.7574 $\pm$ 0.0400	0.0620 $\pm$ 0.0147
pixel_downsample_5pct	8.4340 $\pm$ 1.5407	2.2468 $\pm$ 0.8251	2.5309 $\pm$ 0.2111	0.7665 $\pm$ 0.0382	0.0622 $\pm$ 0.0147
downsample_10pct	8.6884 $\pm$ 1.7005	2.3288 $\pm$ 0.9098	2.5866 $\pm$ 0.2215	0.7574 $\pm$ 0.0397	0.0620 $\pm$ 0.0147
pixel_downsample_10pct	8.4308 $\pm$ 1.5376	2.2362 $\pm$ 0.8280	2.5284 $\pm$ 0.2178	0.7670 $\pm$ 0.0388	0.0621 $\pm$ 0.0150
	8.7053 $\pm$	2.3396 $\pm$	2.5793 $\pm$	0.7591	0.0620 $\pm$

downsample_20pct transform	tangent trace	tangent $\lambda_{\max}$	spectral entropy	$\pm$ top-10 share	circular variance
pixel_downsample_20pct	8.3116 $\pm$ 1.5453	2.2106 $\pm$ 0.8267	2.5163 $\pm$ 0.2115	0.7676 $\pm$ 0.0384	0.0615 $\pm$ 0.0151
downsample_40pct	8.8511 $\pm$ 1.6547	2.3567 $\pm$ 0.9101	2.5755 $\pm$ 0.2080	0.7617 $\pm$ 0.0360	0.0625 $\pm$ 0.0144
pixel_downsample_40pct	8.1782 $\pm$ 1.5154	2.1220 $\pm$ 0.7718	2.4896 $\pm$ 0.2002	0.7729 $\pm$ 0.0358	0.0608 $\pm$ 0.0148
downsample_60pct	8.9885 $\pm$ 1.5274	2.3026 $\pm$ 0.8298	2.5678 $\pm$ 0.1983	0.7650 $\pm$ 0.0341	0.0626 $\pm$ 0.0141
pixel_downsample_60pct	8.0991 $\pm$ 1.4896	2.0949 $\pm$ 0.7705	2.4697 $\pm$ 0.2094	0.7765 $\pm$ 0.0361	0.0588 $\pm$ 0.0142
downsample_80pct	9.0825 $\pm$ 1.4530	2.2022 $\pm$ 0.7401	2.5403 $\pm$ 0.2095	0.7721 $\pm$ 0.0330	0.0598 $\pm$ 0.0141
pixel_downsample_80pct	7.8798 $\pm$ 1.4360	2.0334 $\pm$ 0.7217	2.3964 $\pm$ 0.2044	0.7866 $\pm$ 0.0350	0.0553 $\pm$ 0.0138
downsample_85pct	8.8670 $\pm$ 1.4370	2.1181 $\pm$ 0.7449	2.5217 $\pm$ 0.2287	0.7726 $\pm$ 0.0357	0.0585 $\pm$ 0.0141
pixel_downsample_85pct	7.6474 $\pm$ 1.3430	1.9836 $\pm$ 0.6982	2.3743 $\pm$ 0.2179	0.7871 $\pm$ 0.0360	0.0542 $\pm$ 0.0134
downsample_90pct	8.6496 $\pm$ 1.3900	2.1075 $\pm$ 0.7272	2.5033 $\pm$ 0.2229	0.7787 $\pm$ 0.0332	0.0573 $\pm$ 0.0138
pixel_downsample_90pct	7.7745 $\pm$ 1.3452	2.0491 $\pm$ 0.6944	2.3728 $\pm$ 0.2128	0.7882 $\pm$ 0.0367	0.0547 $\pm$ 0.0128
downsample_93pct	8.2983 $\pm$ 1.4501	2.0909 $\pm$ 0.7254	2.4687 $\pm$ 0.2265	0.7821 $\pm$ 0.0337	0.0577 $\pm$ 0.0140
pixel_downsample_93pct	7.8503 $\pm$ 1.2907	2.1015 $\pm$ 0.6964	2.4103 $\pm$ 0.2141	0.7830 $\pm$ 0.0370	0.0558 $\pm$ 0.0130

Figure 32. Trace by viewpoint angle. Flank viewpoints exhibit the largest trace gain once detail is stripped, pointing to orientation-sensitive uncertainty.

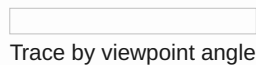


Figure 33. PCA embeddings — Original (baseline). Baseline embeddings remain compact with minimal directional bias.



Figure 34. PCA embeddings — gaussian_noise_500 (noise max trace). Gaussian noise with $\sigma=0.50$ widens the cluster without shifting its centre markedly.

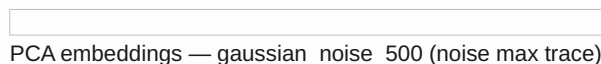


Figure 35. PCA embeddings — gaussian_noise_050 (noise min trace). Gaussian noise with $\sigma=0.05$ widens the cluster without shifting its centre markedly.

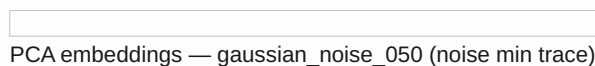


Figure 36. PCA embeddings — `downsample_80pct` (downsample max trace). Smoothed crop (bicubic upsample) at 80% reduction elongates the embedding cloud along a single axis as detail is lost.



Figure 37. PCA embeddings — `downsample_1pct` (downsample min trace). Smoothed crop (bicubic upsample) at 1% reduction elongates the embedding cloud along a single axis as detail is lost.



Figure 38. PCA embeddings — `pixel_downsample_80pct` (pixelated max trace). Pixelation at 80% produces discrete clusters as block artefacts dominate token activations.

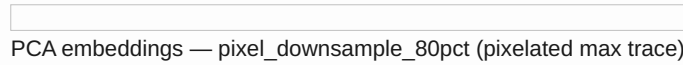
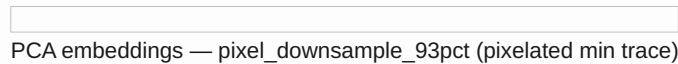


Figure 39. PCA embeddings — `pixel_downsample_93pct` (pixelated min trace). Pixelation at 93% produces discrete clusters as block artefacts dominate token activations.



11. Pass Count Stability

We evaluate trace and log-determinant stability across Monte Carlo pass counts $T \in \{2, 4, 8, 16, 32, 64, 128\}$. Lower T increases estimator noise; curves should flatten as T grows. See stability plots in this section if generated.

Figure 40. Trace stability vs passes. Trace spans shrink to 6.637 for the baseline sweep, while the harsh downsample stays within 5.915 once $T \geq 32$.

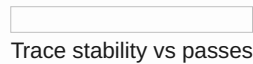
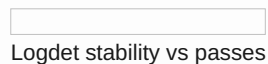


Figure 41. Logdet stability vs passes. Logdet variance collapses as passes double, confirming that 64 draws are ample for stable volume estimates.



12. Detection-to-Embedding Pipeline (YOLOv8 Crop) and Camera Distance

Object scale shrinks as the camera moves away, mixing background into the crop and reducing effective resolution. To benchmark a practical workflow, we introduce a detection → crop → CLIP pipeline and study how distance-driven scale changes alter the embedding cloud.

12.1 Pipeline

- Detect vehicles in the full-resolution frame with YOLOv8 (COCO classes: car, bus, truck).
- Crop the predicted box, optionally expanding by $\approx 5\%$ for context while staying inside image bounds.
- Resize the crop to CLIP's 224×224 input via either bicubic antialiased resampling or nearest-neighbour upsampling.
- Apply CLIP preprocessing and run MCDO ($T=64$, $p=0.01$) to estimate the stochastic embedding cloud.
- Compare crop embeddings to a reference (full frame or near-distance crop)

using mean and Mahalanobis shift, trace/logdet, and anisotropy.

Notes: - CLIP always consumes 224×224 inputs; “raw low-res” crops therefore require an upsample stage. Nearest-neighbour preserves the block grid, while bicubic smooths detail. - Tiny crops widen trace and contract logdet, mirroring the high-severity downsampling response documented above.

12.2 Distance Effects (Expected/Observed)

- Smaller crops (greater distance) increase trace and Mahalanobis shift while logdet contracts.
- Mahalanobis shift reacts faster than raw mean shift because covariance volume shrinks as detail disappears.
- Raw (nearest) crops inject stronger anisotropy and larger shifts than smoothed (bicubic) crops at the same scale.

12.3 Practical Guidance

- Prefer bicubic antialiased resize before sending crops to CLIP; nearest-neighbour magnifies block artefacts.
- Maintain a minimum crop size by padding boxes with a narrow context band before resizing.
- Bin detections by object scale or distance and report mean/Mahalanobis shift, trace, logdet, and anisotropy per bin to visualise degradation.

13. Discussion & Outlook

- **High noise ($\sigma=0.50$)** expands trace by +4.0% and drops logdet by 7.89. The stochastic mean drifts 8.70 L2 units from the deterministic embedding, and tangent variance settles around 7.74, confirming broader but directional uncertainty.
- **Extreme downsampling (93% → 16px)** shifts trace by +4.2% and drops logdet by 4.84. Mahalanobis drift reaches 5948.2, while spectral entropy averages 2.47, signalling variance concentration into fewer modes.
- **Viewpoint sensitivity:** At 225°, trace shifts by +1.38 relative to the clean view, indicating that certain orientations (e.g. flank perspectives) become the most uncertain once detail is removed.
- **Pass-count stability:** Trace estimates converge within ± 3.319 by 32 passes; high-noise and heavy-downsample settings widen the band only modestly, so T=64 remains a safe budget.
- **Predictive head:** Mutual information stays near $2.54e-05$ for all conditions, so predictive entropy contributes little insight compared with embedding-space diagnostics.

14. Class-Level Trace Shifts

class	trace (original)	trace (saltpepper_5pct)	Δ noise	trace (downsample_93pct)	dow
Indigo	11.3669 ± 1.6120	11.7008 ± 1.3592	0.3339	12.2583 ± 1.2720	0.89%
Magenta	11.9280 ± 3.2723	12.0372 ± 3.4081	0.1091	12.6884 ± 3.7444	0.76%
Moose3	12.6069 ± 3.3034	14.2672 ± 3.0083	1.6603	13.4894 ± 3.3945	0.88%
White	13.1045 ± 2.9819	12.9342 ± 2.4708	-0.1703	13.2767 ± 2.5596	0.17%
Wolf2	11.7160 ± 2.8515	12.8059 ± 2.5326	1.0899	12.2501 ± 2.4074	0.53%
Yellow	13.2319 ± 2.3014	13.0068 ± 2.1829	-0.2251	13.1653 ± 2.5896	-0.06%

15. Example Perturbations

Preview grids illustrate how each perturbation family reshapes the rendered jeep: baseline is shown alongside rising severities.

Figure 42. Jeep previews — noise sweep. Gaussian noise progressively speckles the frame while salt & pepper corruption introduces isolated extreme pixels.



Figure 43. Jeep previews — smoothed crop sweep. Smoothing the coarse crop blurs structure as resolution falls, culminating in a soft, low-detail silhouette.

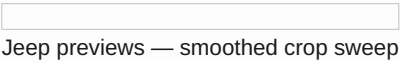
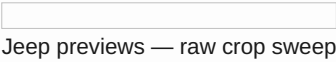


Figure 44. Jeep previews — raw crop sweep. Nearest-neighbour reductions replace detail with large blocks, emphasising aliasing artefacts at coarse grids.



Appendix: Detailed Metrics

Aggregate metrics

transform	trace	logdet	anisotropy (corr-F)
original	12.3135 ± 2.6974	-6376.0333 ± 10.0092	169.6903 ± 19.4235
gaussian_noise_010	12.6023 ± 2.7577	-6374.4188 ± 9.8618	169.3933 ± 18.9273
gaussian_noise_020	12.5631 ± 2.6985	-6373.7425 ± 9.3996	165.5048 ± 18.6221
gaussian_noise_050	12.3926 ± 2.6261	-6375.8614 ± 8.2435	166.7742 ± 18.1294
gaussian_noise_100	12.4558 ± 2.5969	-6376.6802 ± 7.1463	167.6568 ± 18.4162
gaussian_noise_200	12.5629 ± 2.5728	-6377.9926 ± 6.7516	166.6415 ± 19.0206
gaussian_noise_500	12.8031 ± 2.7534	-6383.9255 ± 6.6858	166.3067 ± 18.1931
saltpepper_5pct	12.7279 ± 2.5296	-6375.3044 ± 6.3954	163.6120 ± 18.6287
downsample_1pct	12.3825 ± 2.6268	-6375.9890 ± 9.1654	167.8046 ± 18.3878
pixel_downsample_1pct	12.5172 ± 2.6165	-6377.2888 ± 8.4542	168.4906 ± 18.3043
downsample_2pct	12.3977 ± 2.6292	-6375.7912 ± 9.2649	167.7650 ± 18.2637
pixel_downsample_2pct	12.5455 ± 2.6315	-6376.9582 ± 8.7136	167.9582 ± 18.1520
downsample_5pct	12.4478 ± 2.6403	-6375.7133 ± 9.0908	167.4770 ± 18.0066
pixel_downsample_5pct	12.5134 ± 2.5646	-6377.1219 ± 8.2152	168.4439 ± 17.7280
downsample_10pct	12.4398 ± 2.6474	-6375.7898 ± 9.0936	167.4970 ± 18.0020
pixel_downsample_10pct	12.5616 ± 2.6045	-6377.0398 ± 8.0190	168.4702 ± 18.0116
downsample_20pct	12.4790 ± 2.6444	-6375.9428 ± 9.0444	167.5528 ± 18.0000

	transform	trace	logdet	anisotropy (GSI)
pixel_downsample_20pct	2.6145 ± 12.4631 ± 2.5723	8.7840 ± -6377.9056 ± 7.6650	17.9934 ± 169.1283 ± 17.8052	
downsample_40pct	12.6968 ± 2.5948	-6375.4607 ± 8.3049	166.6609 ± 18.0419	
pixel_downsample_40pct	12.4773 ± 2.5321	-6378.9127 ± 8.2044	170.3822 ± 17.2721	
downsample_60pct	13.0259 ± 2.5301	-6374.7296 ± 7.5273	165.0812 ± 17.6869	
pixel_downsample_60pct	12.5031 ± 2.5833	-6380.0696 ± 7.6918	169.9147 ± 17.9723	
downsample_80pct	13.5591 ± 2.5690	-6374.3342 ± 6.5821	161.3558 ± 17.6353	
pixel_downsample_80pct	12.6936 ± 2.6429	-6382.2148 ± 7.4643	163.9830 ± 17.1538	
downsample_85pct	13.5157 ± 2.6575	-6375.1753 ± 6.4177	161.0859 ± 18.7373	
pixel_downsample_85pct	12.4749 ± 2.5475	-6383.4985 ± 7.2516	161.6255 ± 17.7324	
downsample_90pct	13.2171 ± 2.5903	-6377.9928 ± 6.2483	159.3951 ± 17.3251	
pixel_downsample_90pct	12.5712 ± 2.5332	-6383.3493 ± 6.4484	161.4009 ± 16.0536	
downsample_93pct	12.8271 ± 2.6290	-6380.8724 ± 6.6015	161.2282 ± 16.4835	
pixel_downsample_93pct	12.3748 ± 2.4221	-6382.7461 ± 6.0468	163.5343 ± 15.9229	

Noise severity deltas

transform	Δ trace (%)	Δ logdet	Δ off-diag (%)
gaussian_noise_010	2.35	1.6145	1.71
gaussian_noise_020	2.03	2.2908	-1.95
gaussian_noise_050	0.64	0.1719	-3.46
gaussian_noise_100	1.16	-0.6468	-3.42
gaussian_noise_200	2.03	-1.9593	-5.86
gaussian_noise_500	3.98	-7.8921	-11.82
saltpepper_5pct	3.37	0.7289	-4.74

Smoothed crop severity deltas

transform	Δ trace (%)	Δ logdet	Δ off-diag (%)
downsample_1pct	0.56	0.0444	-1.76
downsample_2pct	0.68	0.2421	-1.68
downsample_5pct	1.09	0.3200	-1.61
downsample_10pct	1.03	0.2436	-1.62
downsample_20pct	1.34	0.0905	-1.52
downsample_40pct	3.11	0.5727	-1.14
downsample_60pct	5.79	1.3038	-0.81
downsample_80pct	10.12	1.6991	-3.16
downsample_85pct	9.76	0.8580	-4.94
downsample_90pct	7.34	-1.9594	-10.22
downsample_93pct	4.17	-4.8391	-12.43

Raw crop severity deltas

transform	Δ trace (%)	Δ logdet	Δ off-diag (%)
pixel_downsample_1pct	1.65	-1.2554	-1.62
pixel_downsample_2pct	1.88	-0.9249	-1.73
pixel_downsample_5pct	1.62	-1.0885	-1.63
pixel_downsample_10pct	2.02	-1.0065	-1.52
pixel_downsample_20pct	1.22	-1.7722	-2.08
pixel_downsample_40pct	1.33	-2.8794	-2.58
pixel_downsample_60pct	1.54	-4.0363	-4.44
pixel_downsample_80pct	3.09	-6.1815	-11.66
pixel_downsample_85pct	1.31	-7.4652	-16.07
pixel_downsample_90pct	2.09	-7.3160	-15.49
pixel_downsample_93pct	0.50	-6.7128	-13.23

YOLOv8 detection robustness vs downsampling

Using the Section 12 pipeline (YOLOv8n detection → crop → CLIP resize), we evaluated detections for every original frame and each smoothed (bicubic) / raw (nearest) crop variant. Vehicle detections cover COCO classes {car, bus, truck}. Detailed outputs live in runs/sim2_noise_study/yolov8/variant_summary.csv and the accompanying per-image CSV. - Baseline detection succeeds on 43/46 views (93.5%), leaving 3 hard cases. - Smoothed crop (bicubic) recall trends: 95.7% at 40% reduction, 82.6% at 60% reduction, 67.4% at 80% reduction, 21.7% at 90% reduction, and 4.3% at 93% reduction. - Raw crop (nearest) recall drops to 82.6% at 10% reduction, 69.6% at 20% reduction, 54.3% at 40% reduction, and 2.2% at 60% reduction; detections vanish once reductions exceed 80%. - Confidence slips from 0.61 (original) to 0.22 at 90% smoothed crop while boxes still span 94.1% → 90.2% of the frame.

variant	detection rate (%)	mean best conf	mean box area (%)
original	93.5	0.61	94.1
downsample_1pct	95.7	0.61	91.4
downsample_2pct	93.5	0.63	93.6
downsample_5pct	95.7	0.63	91.3
downsample_10pct	93.5	0.62	91.2
downsample_20pct	95.7	0.61	89.5
downsample_40pct	95.7	0.61	89.3
downsample_60pct	82.6	0.56	87.3
downsample_80pct	67.4	0.46	85.9
downsample_85pct	54.3	0.34	80.9
downsample_90pct	21.7	0.22	90.2
downsample_93pct	4.35	0.13	91.2
pixel_downsample_1pct	84.8	0.60	88.1
pixel_downsample_2pct	84.8	0.59	86.1
pixel_downsample_5pct	84.8	0.54	85.2
pixel_downsample_10pct	82.6	0.50	74.0
pixel_downsample_20pct	69.6	0.49	62.8
pixel_downsample_40pct	54.3	0.33	36.7
pixel_downsample_60pct	2.17	0.17	45.0
pixel_downsample_80pct	0.00	—	—
pixel_downsample_85pct	0.00	—	—
pixel_downsample_90pct	0.00	—	—
pixel_downsample_93pct	0.00	—	—

Appendix: Predictive Diagnostics (MI & Entropy)

Predictive mutual information (epistemic) and entropy are derived from the CLIP

text head over class prompts. They remain near-zero here due to prompt unanimity and low dropout; we include them for completeness.

Figure 45. Mutual information across transforms. Predictive mutual information remains near zero for every perturbation, confirming the head stays confident despite dropout.



Mutual information across transforms

Figure 46. Entropy across transforms. Predictive entropy barely moves across conditions, reinforcing that embedding diagnostics carry the informative signal.



Entropy across transforms